INCH-POUND
MIL-PRF-83866(USAF)
w/Amendment 1
1 SEPTEMBER 2025
SUPERSEDING
MIL-R-83866(USAF)
19 JULY 1974

PERFORMANCE SPECIFICATION

RECEIVER, OXYGEN MASK, ASSEMBLY OF

MIL-PRF-83866(USAF) w/Amendment 1 remains inactive for new design, dated 10 Oct 1997 and is no longer used, except for replacement purposes.

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

This specification covers the oxygen mask receiver assembly used to attach the MBU-12/P and MBU-20/P oxygen mask to the protective helmet assembly for aircrew members.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB OH 45433-7017 or e-mailed to Engineering.Standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 1660

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-P-1757	Primer Coating, Alkyd Base, One Component

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MS35335	Washer, Lock, Flat-External Tooth
MS51957	Screw, Machine-Pan Head, Cross Recessed, Corrosion-Resisting Steel UNC-2A

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-831	Preparation of Test Reports

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS – AIR FORCE

60B4462	Plate, Mounting, Receiver Jaw
60B4464	Plate-Mounting, Receiver Jaw, Oxygen Mask
60B4465	Back Plate, Housing-Oxygen Mask Receiver
60B4466	Nameplate-Receiver, Oxygen Mask
60C4459	Receiver, Oxygen Mask, Assembly of
60C4461	Jaw, Receiver, Oxygen Mask, Assembly of
60C4463	Jaw-Receiver, Oxygen Mask
62A4257	Rivet, Tubular, Receiver Jaw
70280	Bayonet, Offset, Double Eyelet-Oxygen Mask

(Copies of these drawings are available online at https://a-team.us.af.mil/login/, for users with approved access rights granted through the All-purpose Tool for Engineering, Analytics, and Management (A-TEAM). Users not located on a *.mil network may contact the A-TEAM Help Desk via email at wralc.tila.centra@us.af.mil for instructions on obtaining and completing the necessary forms.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications.
The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA (AIA)/NATIONAL AEROSPACE STANDARD (NAS)

NASM122116-122155	Insert, CRES, Helical Coil, Coarse Thread, 1-1/2 Dia Nominal Length (DoD adopted)

(Copies of this document are available online at https://store.accuristech.com/.)

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.24	Types and Applications of Engineering (DoD adopted)
ASME Y14.34	Associated Lists (DoD adopted)
ASME Y14.35	Revision of Engineering Drawings and Associated Documents (DoD adopted)
ASME Y14.100	Engineering Drawing Practices (DoD adopted)

(Copies of these documents are available online at https://www.asme.org/.)

ASTM INTERNATIONAL

ASTM B26/B26M	Standard Specification for Aluminum-Alloy Sand Castings (DoD adopted)
ASTM B85/B85M	Standard Specification for Aluminum-Alloy Die Castings (DoD adopted)
ASTM B108/B108M	Standard Specification for Aluminum-Alloy Permanent Mold Castings (DoD adopted)
ASTM B134/B134M	Standard Specification for Brass Wire
ASTM B159/B159M	Standard Specification for Phosphor Bronze Wire (DoD adopted)
ASTM B206/B206M	Standard Specification for Copper-Nickel (Nickel Silver) Wire and Copper-Nickel Alloy Wire (DoD adopted)
ASTM B209/B209M	Standard Specification for Aluminum and Aluminum Alloy Sheet and Plate (DoD adopted)
ASTM B783	Standard Specification for Materials for Ferrous Powder Metallurgy (PM) Structural Parts

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE AMS4290	Aluminum Alloy, Die Castings 9.5Si – 0.50Mg (360.0-F) As Cast – UNS A03600
SAE AMS4291	Aluminum Alloy, Die Castings 8.5Si – 3.5Cu (A380.0-F) As Cast – UNS A13800
SAE AMS5513	Steel, Corrosion-Resistant, Sheet, Strip, and Plate 19Cr – 9.2Ni (SAE 30304) Solution Heat Treated – UNS S30400

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

SAE AMS5516	Steel, Corrosion-Resistant, Sheet, Strip, and Plate18Cr – 9.0Ni (302) Solution Heat Treated – UNS S30200
SAE AMS5517	Steel, Corrosion Resistant, Sheet and Strip 18Cr - 8Ni (301) Cold Rolled, 125 ksi (862 MPa) Tensile Strength - UNS S30100
SAE AMS5518	Steel, Corrosion Resistant, Sheet and Strip 18Cr - 8Ni Cold Rolled, 1/2 Hard, 150 ksi (1034 MPa) Tensile Strength - UNS S30100
SAE AMS5519	Steel, Corrosion Resistant, Sheet and Strip 18Cr – 8Ni (SAE 30301) Cold Rolled, Full Hard, 185 ksi (1276 MPa) Tensile Strength - UNS S30100
SAE AMS5524	Steel, Corrosion and Heat-Resistant, Sheet, Strip and Plate 18Cr – 13Ni – 2.5Mo (SAE 30316) Solution Heat Treated – UNS S31600
SAE AMS5901	Steel, Corrosion-Resistant, Sheet, Strip, and Plate 18Cr - 8Ni (SAE 30301) Solution Heat Treated - UNS S30100
SAE AMS5902	Steel, Corrosion-Resistant, Sheet and Strip 18Cr – 8Ni (301) Cold Rolled, 3/4 Hard, 175 ksi (1207 MPa) Tensile Strength - UNS S30100
SAE AMS5903	Steel, Corrosion-Resistant, Sheet, Strip, and Plate 18Cr - 9.0Ni (302) Cold Rolled, 1/4 Hard, 125 ksi (862 MPa) Tensile Strength - UNS S30200
SAE AMS5904	Steel, Corrosion-Resistant, Sheet and Strip 18Cr - 9.0Ni (302) Cold Rolled, 1/2 Hard, 150 ksi (1034 MPa) Tensile Strength - UNS S30200
SAE AMS5905	Steel, Corrosion Resistant, Sheet and Strip 18Cr - 9.0Ni (302) Cold Rolled, 3/4 Hard, 175 ksi (1207 MPa) Tensile Strength - UNS S30200
SAE AMS5906	Steel, Corrosion Resistant, Sheet Strip 18Cr - 9.0Ni (SAE 30302) Cold Rolled, Full Hard, 185 ksi (1276 MPa) Tensile Strength - UNS S30200
SAE AMS5907	Steel, Corrosion and Heat Resistant, Sheet, Strip, and Plate 17Cr - 12Ni - 2.5Mo (316) Cold Rolled, 1/4 Hard, 125 ksi (862 MPa) Tensile Strength - UNS S31600
SAE AMS5910	Steel, Corrosion-Resistant, Sheet, Strip, and Plate 19Cr - 9.2Ni (304) Cold Rolled, 1/4 Hard, 125 ksi (862 MPa) Tensile Strength - UNS S30400
SAE AMS5911	Steel, Corrosion-Resistant, Sheet and Strip 19Cr - 9.2Ni (304) Cold Rolled, 1/2 Hard, 150 ksi (1034 MPa) Tensile Strength - UNS S30400
SAE AMS5912	Steel, Corrosion-Resistant, Sheet and Strip 19Cr - 9.2Ni (304) Cold Rolled, 3/4 Hard, 175 ksi (1207 MPa) Tensile Strength - UNS S30400
SAE AMS5913	Steel, Corrosion-Resistant, Sheet and Strip 19Cr - 9.2Ni (304) Cold Rolled, Full Hard, 185 ksi (1276 MPa) Tensile Strength - UNS S30400
SAE-AMS-QQ-A-250	Aluminum and Aluminum Alloy, Plate and Sheet General

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

	Specification for
SAE J452	General Information – Chemical Compositions, Chemical Compositions, Mechanical and Physical Properties of SAE Aluminum Casting Alloys

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence.
Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification.
The receiver assembly furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Materials.
Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Metals.
Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to occur in storage or normal service. Metals shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective treatment that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.3 Design and construction.
The oxygen mask receiver assembly shall be complete when the receiver jaw assembly is in position within the receiver housing and the nameplate attached to the top of the housing with lock washers, MS35335-58, and machine screws, MS51957-28. Also, the backplate must be attached to the housing with lock washers, MS35335-58, and machine screws MS51957-26 and MS51957-28. The receiver assembly shall conform to AF Drawing 60C4459.

3.3.1 Nameplate.
The nameplate shall be made of aluminum alloy, temper T4, 0.020 inches thick, in accordance with SAE-AMS-QQ-A-250, ASTM B209/B209M and shall conform to AF Drawing 60B4466.

3.3.2 Backplate.
The backplate shall be made of stainless steel, composition 301, 1/4 hard, in accordance with SAE AMS5513, SAE AMS5910 thru 5913, SAE AMS5524, SAE AMS5516 thru 5519, SAE AMS5901 thru 5907 and shall conform to AF Drawing 60B4465.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

3.3.3 Jaw assembly.
The jaw assembly shall be assembled in accordance with AF Drawing 60C4461. The jaw assembly is composed of a top mounting plate, AF Drawing 60B4462; a bottom mounting plate, AF Drawing 60B4464; jaws, AF Drawing 60C4463; and rivets, AF Drawing 62A4257.

3.3.3.1 Mounting plates.
Both mounting plates shall be made of stainless steel, composition 301, 1/4 hard in accordance with SAE AMS5513, SAE AMS5910 thru 5913, SAE AMS5524, SAE AMS5516 thru 5519, SAE AMS5901 thru 5907 and shall conform to AF Drawings 60B4462 and 60B4464.

3.3.3.2 Jaws.
The jaws shall be made of copper infiltrated sintered carbon steel, Class C in accordance with ASTM B783.

3.3.3.3 Rivets.
The rivets shall be made of copper alloy, alloy number 260, temper 1/2 hard in accordance with ASTM B134/B134M, ASTM B159/B159M, ASTM B206/B206M. The rivets shall conform to AF Drawing 62A4257.

3.4 Performance.

3.4.1 Receiver assembly.
The receiver assembly, when fully prepared according to this specification, shall pass the visual examination (see 4.5.1) and the pull test (see 4.5.2).

3.5 Interchangeability.
All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of ASME Y14.100, ASME Y14.24, ASME Y14.34, ASME Y14.35 shall govern changes in the manufacturer's part numbers.

3.6 Identification of product.
The receiver assembly shall have the following information impregnated or stamped on the nameplate surface in accordance with AF Drawing 60B4466 and MIL-STD-130.

Manufacturer's Name or Code (Manufacturer shall include the applicable information).
P/N 60C4459
Contract No (Manufacturer shall include the applicable information).

3.7 Workmanship.
The receiver assembly shall be uniform in quality and shall be free from irregularities, defects or foreign matter which could adversely affect safety, performance, reliability, or durability.

4. VERIFICATION

4.1 Classification of inspection.
The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

4.2 Responsibility for inspection.
Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.3 Qualification inspection.
Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products.

4.3.1 Qualification samples.
Three receiver assemblies shall be prepared in accordance with this specification and shall be submitted to the procuring activity for qualification. The receiver assembly shall be representative of the production item.

4.3.2 Qualification test report.
The contractor shall prepare three copies of a test report in accordance with MIL-HDBK-831 and submit it to the qualifying activity. The test report shall contain detailed statements, identified by the applicable paragraph number of this specification, of compliance or non-compliance with each requirement of this specification (see 6.2).

4.4 Conformance inspection.
Conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.4.1 Individual tests.
Each receiver assembly shall be visually examined in accordance with 4.5.1.

4.4.2 Sampling tests.
Three receiver assemblies selected at random from each lot of 100 or less shall be subjected to the pull test in accordance with 4.5.2.

4.4.2.1 Rejection and retest.
When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.4.2.2 Individual tests may continue.
For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.4.3 Defects in items already accepted.
The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

4.5 Test methods.

4.5.1 Visual examination.

The receiver assembly shall be visually examined according to Table I. One major defect or ten minor defects out of a lot shall constitute rejection.

TABLE I. Classification of defects for visual examination.

MAJOR	MINOR
1. Dimensions not within tolerance limits.	101. Identification is illegible or missing.
2. Screws missing or threads stripped.	102. Screws are not tight or securely attached to the receiver.
3. Jaw assembly not in proper position within housing.	
4. Any material not conforming to specification.	
5. Any part or component missing from receiver assembly.	
6. Any defect which hinders the function of the receiver assembly.	

4.5.2 Pull test.

The receiver assembly with a bayonet, in accordance with AF Drawing 70280, inserted shall withstand a maximum separating force of 500 pounds without damage to the receiver. A pull force shall be applied to the receiver engagement, while the bayonet is being locked in a plane coinciding with the receiver jaw assembly, in increments of 25 pounds until the force reaches 200 pounds. The receiver shall then be subjected to a pull force of 500 pounds and the connection shall not fail. Any damage or distortion to the receiver assembly during any part of the test constitutes rejection.

5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

The receiver assembly is intended to be used as an attachment to the aircrew member's helmet assembly and use the oxygen mask bayonet assembly as its mating connection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements.
Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. List of interchangeable components and assemblies (see 3.4).
c. Copy of QPL test report (see 4.3.2).
d. Packaging requirements (see 5.1).

6.3 Qualification.
With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 83866 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFLCMC/ROU Bldg. 300 West Wing, Bay C, Rm C314, Robins AFB GA, 31098 or emailed to Engineering.Standards@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.
Bayonet
Jaw receiver
Mounting plate

6.5 Amendment notations.
The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the content irrespective of the marginal notations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Air Force - 184

Preparing activity:
Air Force - 184

Reviewer:
DLA - GS

Agent:
Air Force - 110

(Project 1660-2025-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.